P2 - CSG - Sheet and Print

ARC 3133-001 / Architectural Visualization Methods 1 / Fall 2026 - Full Term

Approach: Procedural

Revision: 2026-11-02. Current assignment requirements and deadlines. Completed midterm work is reviewed using the evidence already submitted; no retrospective deliverables are added.

2.1 - CSG - an architectural object as instructions

Introduced: Week 2 - Sep 1 / **Due:** Sun Oct 18, 11:59 PM Eastern Time / **Weight:** 10%

Purpose

Reinterpret the main form of a real architectural project as a short, understandable sequence of solid operations.

What you will learn

- Navigate the basic Geometry Nodes or Grasshopper interface: connect nodes, identify inputs and outputs, and change a value.
- Move, rotate and scale geometry; understand union, subtraction and intersection.
- Break an architectural object into an ordered list of instructions that someone else can follow.

What to do

- Choose a documented real project. Credit the project, architect and image source; identify the main forms you will simplify.
- Build an editable model from basic solids and transformations. Demonstrate union, subtraction and intersection; a small supporting comparison is acceptable if one operation is unnecessary in the main object.
- Explain the construction in about six to nine diagram steps. Match each step to a short instruction in ordinary language; formal code is not required.
- Use BIG-style graphics: white building masses, restrained color for changes or operations, clear outlines and simple line weights. Show context or vegetation in the final view.
- Name or separate objects when it helps explain the sequence. Use the BIG-style diagram tutorial in Resources.

What to submit

- **Presentation:** one 17 x 11 inch landscape sheet, `Lastname_Firstname_P2a.pdf`.
- **On the sheet:** a small credited precedent image; about six to nine numbered construction diagrams; matching numbered instructions; and a final architectural view with context. Keep each instruction beside its diagram.
- **Operations:** make the transformations and Boolean operations legible. If an operation is demonstrated outside the main sequence, place the small comparison on this same sheet.
- **Editable model:** include `.blend`, or `.gh` with its referenced `.3dm`, plus external dependencies. Keep the construction steps and useful controls accessible.
- **Package:** `Lastname_Firstname_P2a/`. Open the model from the submission folder and confirm it reproduces the final view. Instructions may be plain language; no separate code document is required.

What we assess

- Can you explain and change the construction sequence? Are the transformations and Boolean operations used intentionally? Does the sheet communicate the architectural interpretation clearly?
- Digital rubric: understanding 30%, communication 30%, technical execution 25%, exploration 15%.

2.2 - 3D printing - from model to layers

Introduced: Week 3 - Sep 8 / **Due:** Sun Oct 18, 11:59 PM Eastern Time / **Weight:** 5%

Preparation checks: Week 6 - Sep 29: Before you queue the print, check the geometry, units, orientation and sliced preview.

Purpose

Prepare and print your CSG object, using the slicer to understand how the machine will build it.

What you will learn

- Recognize the slicer workspace, object placement, settings, supports, estimates and layer preview.
- Understand layers, walls, infill, adhesion, orientation, overhangs and bridging.
- Connect common print problems to a setting or geometry decision.

What to do

- Check units and size, choose an orientation and inspect the layer preview before printing.
- Use a strategy to avoid or reduce supports: reorient the object, separate it into parts or adjust overhangs. Bridging is acceptable when the test or preview supports it.
- Identify the settings that matter for your object: layer height, walls, infill, adhesion and supports. Explain your choices briefly rather than listing every software option.
- Keep the assembled object within 60 mm on each axis and within the existing 200 g filament allowance, including tests and failed attempts.
- Document a problem and response, or an anticipated problem checked in the preview. Follow the FabLab workflow.

What to submit

- **Physical outcome:** one assembled CSG print within 60 mm on each axis. The recorded midterm submission is an upload of print photographs to Teams, not a new physical hand-in.
- **Documentation:** `Lastname_Firstname_P2b.pdf` with an overall photograph of the finished object, an annotated slicer setup view and an annotated layer-preview view. Add a detail or failure photograph when it helps explain the result.
- **Explain briefly:** object dimensions; orientation or part splitting; layer height, walls, infill, adhesion and support choices; and one problem/test and your response. Use existing CSG documentation or the course template; no additional polished process sheet is required.
- **Machine/source files:** editable geometry, exported print mesh (`.stl` or `.3mf`) and the saved slicer project where supported; otherwise include the settings in the documentation.
- **Package:** `Lastname_Firstname_P2b/`. Check units, exported dimensions and that the documented settings match the printed result. Do not submit only a mesh or only photographs.

What we assess

- Preparation and fabrication strategy 30%; fabrication quality 30%; visual documentation 25%; testing and revision 15%. Explain what the slicer shows and what you learned from the result.

Shared standards: Apply the GSM and credit sources. Package linked assets; check PDFs and editable files before uploading. Explain rules plainly and follow the course submission policies.